MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER TYPE 2N398A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification dovers the detail requirements for a PNP, germanium, low-power transistor.

1.2 Physical dimensions. See figure 1 (TO-5)

1.3 Maximum ratings.

P_T 1/	V_{CBO}	V_{CES}	V_{EBO}	I_C	T_{stg}
mW	Vdc	Vdc	Vdc	mAdc	°C
150	-105	-105	-75	-200	-65 to +100

1/ Derate linearly 2.0 mW/°C for $T_A > 25°C$.

1.4 Primary electrical characteristics.

Limits	f_{hfb} V_{CB} = -6 Vdc I_C = +1 mAdc	h_{FE} V_{CE} = -0.35 Vdc I_C = -5 mAdc	C_{obo} V_{CB} = -6 Vdc I_E = 0 100 kHz $\leq f \leq$ 1 MHz	V_{CE} (sat) I_C = -100 mAdc I_B = -20 mAdc
	MHz		pF	Vdc
Min	0.6	20	---	---
Max	---	100	25	-0.5

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

R_{BE} - - - - - - - - - - External base to emitter resistance

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

* 3.3.1 Lead material and finish. Lead material shall be Kovar or alloy 52. Lead finish shall be gold-plated. (Leads may be tin-coated if specified in the contract or order, and this requirement shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking (see 6.2)).

* 3.3.1.1 Selectivity of lead material. Where choice of lead material (see 3.3.1 above) is desired, it shall be specified in the contract or order (see 6.2).

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

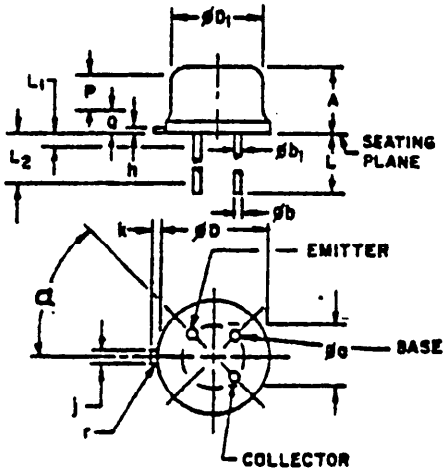
4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every six months during production.

* 4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hour life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria, see 4.3.3.

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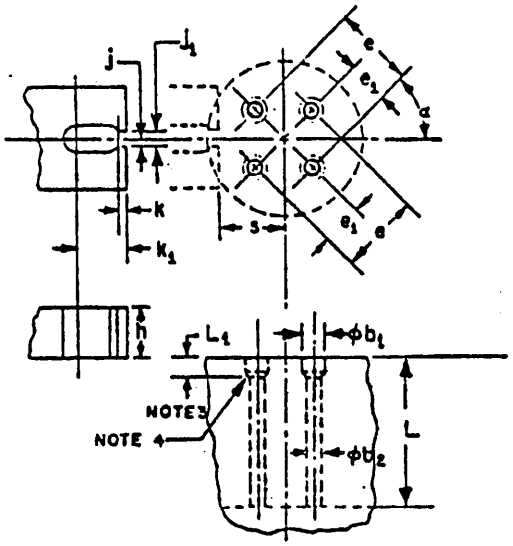


SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.240	.260	6.10	6.60	
øa	.200 TP		5.080 TP		6
øb	.016	.021	.407	.533	7, 8
$øb_1$	.016	.019	.407	.482	7, 8
øD	.335	.370	8.51	9.40	
$øD_1$	.305	.335	7.75	8.51	
h	.009	.041	.23	1.04	
j	.028	.034	.712	.863	2
k	.029	.045	.74	1.14	3
L	1.500	1.750	38.10	44.45	7, 8
L_1	---	.050	---	1.27	7, 8
L_2	.250	---	6.35	---	7, 8
P	.100	---	2.54	---	5
Q	---	.050	---	1.27	4
r	---	.010	---	.254	10
α	45° TP		45° TP		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Beyond r(radius) maximum, j must be held for a minimum length of .011 (.28 mm).
3. k measured from maximum øD.
4. Outline in this zone is not controlled.
5. $øD_1$ shall not vary more than .010 (.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gage plane .054 +.001 -.000 (1.37 +.03 -.00 mm) below seating plane shall be within .007 (.18 mm) radius of True Position at maximum material condition relative to tab at MMC. The device may be measured by direct methods or by the gage and gaging procedure shown in Figure 2.
7. $øb_1$ applies between L_1 and L_2. øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All three leads.
9. All leads electrically insulated from the case.
10. r(radius) applies to both inside corners of tab.

FIGURE 1. Physical dimensions of transistor type 2N398A (TO-5).

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SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ϕb_1	.0595	.0605	1.511	1.537	
ϕb_2	.0325	.0335	.8255	.8509	
e	.1995	.2005	5.067	5.093	
e_1	.0995	.1005	2.527	2.552	
h	.150 NOM		3.81 NOM		
j	.0175	.0180	.4445	.4572	
J_1	.0350	.0355	.8890	.9017	
k	.009	.011	.23	.28	
k_1	.125 NOM		3.18 NOM		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
2. The following gaging procedure shall be used:

   The device being measured shall be inserted until its seating plane is .125(3.18 mm) ± .010(.25 mm) from the seating surface of the gage. A force of 8 ± .5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.

   The use of a pin straightener prior to insertion in the gage is permissible.

3. Gaging plane.
4. Drill angle.

FIGURE 2. Gage for lead and tab location for transistor type 2N398A.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 1							
Visual and mechanical examination	2071		10	---	---	---	---
Subgroup 2							
Breakdown voltage, collector to base	3001	Bias condition D I_C = -50 μAdc	5	BV_{CBO}	-105	---	Vdc
Breakdown voltage, emitter to base	3026	Bias condition D I_E = -100 μAdc		BV_{EBO}	-75	---	Vdc
Collector to base cutoff current	3036	Bias condition D V_{CB} = -60 Vdc		I_{CBO}	---	-14	μAdc
Collector to emitter cutoff current	3041	Bias condition C V_{CE} = -105 Vdc		I_{CES}	---	-600	μAdc
Collector to emitter cutoff current	3041	Bias condition B V_{CE} = -55 Vdc R_{BE} = 10 kilohms		I_{CER}	---	-600	μAdc
Emitter to base cutoff current	3061	Bias condition D V_{EB} = -50 Vdc		I_{EBO}	---	-14	μAdc
Floating potential	3020	V_{CB} = -105 Vdc Voltmeter input resistance ≥ 10 megohms		V_{EBF}	---	-1	Vdc
Subgroup 3							
Forward-current transfer ratio	3076	V_{CE} = -0.35 Vdc I_C = -5 mAdc	5	h_{FE}	20	100	---
Base emitter voltage (saturated)	3066	Test condition A I_C = -5 mAdc I_B = -0.25 mAdc		$V_{BE(sat)}$	---	-0.35	Vdc
Collector to emitter voltage (saturated)	3071	I_C = -5 mAdc I_B = -0.25 mAdc		$V_{CE(sat)}$	---	-0.35	Vdc
Collector to emitter voltage (saturated)	3071	I_C = -100 mAdc I_B = -20 mAdc		$V_{CE(sat)}$	---	-0.5	Vdc
Subgroup 4							
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -6 Vdc I_C = -1 mAdc	5	h_{fe}	20	---	---
Small-signal short-circuit forward-current transfer ratio cutoff frequency	3301	V_{CB} = -6 Vdc I_E = +1 mAdc		f_{hfb}	0.6	---	MHz
Open-circuit output capacitance	3236	V_{CB} = -6 Vdc; I_E = 0 100 kHz ≤ f ≤ 1 MHz		C_{obo}	---	25	pF

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TABLE I. Group A inspection - Continued.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 5			10				
High-temperature operation: Collector to base cutoff current	3036	T_A = +71°C Bias condition D V_{CB} = -60 Vdc		I_{CBO}	---	-200	µAdc
Low-temperature operation: Forward-current transfer ratio	3076	T_A = -55°C V_{CE} = -0.35 Vdc I_C = -5 mAdc		h_{FE}	10	---	---

TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 1			20				
Physical dimensions	2066	See figure 1		---	---	---	---
* Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition B except in step 3, T_A = +100 ±5°C		---	---	---	---
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D V_{CB} = -60 Vdc		I_{CBO}	---	-14	µAdc
Forward-current transfer ratio	3076	V_{CE} = -0.35 Vdc I_C = -5 mAdc		h_{FE}	20	100	---
* Subgroup 3			15				
Shock	2016	Nonoperating; 500 G 1 ms, 5 blows in each orientation: X_1, Y_1, Y_2 and Z_1		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2 and Z_1		---	---	---	---
End points: (Same as subgroup 2)							

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TABLE II. Group B inspection - Continued.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test condition E		---	---	---	---
End points:							
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
* Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
* Subgroup 6			7				
High-temperature life (nonoperating)	1032	T_{stg} = +100°C time = 340 hours (see 4.3.4)		---	---	---	---
End points;							
Collector to base cutoff current	3036	Bias condition D V_{CB} = -60 Vdc		I_{CBO}	---	-25	µAdc
Forward-current transfer ratio	3076	V_{CE} = -0.35 Vdc I_C = -5 mAdc		h_{FE}	15	---	---
* Subgroup 7			7				
Steady-state operation life	1027	P_T = 150 mW V_{CB} = -20 Vdc time = 340 hours (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Barometric pressure, reduced (altitude operation)	1001	Normal mounting; pressure = 8 mm Hg for 60 s min.		---	---	---	---
Measurement during test:							
Collector to base cutoff current	3036	Bias condition D V_{CB} = -105 Vdc		I_{CBO}	---	-50	µAdc

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TABLE III. Group C inspection - Continued.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2			20				
Burnout by pulsing	3005	V_{CB} = 0; I_C = -200 mAdc t_p = 60 s for 1 cycle		---	---	---	---
End points:							
Forward-current transfer ratio	3076	V_{CE} = -0.35 Vdc I_C = -5 mAdc (see 4.4.2)		h_{FE}	20	100	---
* Subgroup 3			10				
Resistance to solvents	---	MIL-STD-202, method 215 (see 4.4.1)		---	---	---	---
* Subgroup 4			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
* Subgroup 5			λ = 10				
Steady-state operation life	1026	P_T = 150 mW V_{CB} = -20 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

* 4.4.1 Resistance to solvents. Transistors shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

4.4.2 Stabilization time. End-point measurements. after the burnout by pulsing test (group C inspection), shall be performed within the time period between 24 hours and 96 hours to allow for stabilization of units.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

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* 6.2 Ordering data.

(a) Lead finish if other than gold-plated (see 3.3.1).
(b) Selectivity of lead material (see 3.3.1.1).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU
Air Force - 11, 80, 70
DSA - ES

User activities:
Army - AV, SM
Navy - AS, CG, MC, SH
Air Force - 13, 15, 19

Preparing activity:
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SPECIFICATION

MIL-S-19500/174B (EC)

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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